Diffusion-Based Generation of Financial Time Series

Primary reference: *Generative Learning for Financial Time Series with Irregular and Scale-Invariant Patterns* (ICLR 2024 Spotlight). OpenReview: PDF, code. ICLR page: link.

Summary. The paper proposes a generative model for financial time series that proceeds in three stages: it extracts representative short segments ("motifs") using a time-warping-type similarity, learns to generate motif segments via a conditional diffusion mechanism, and then assembles long paths by a transition rule. The evaluation focuses on matching stylised facts such as heavy tails and volatility clustering, and on downstream checks that aim to avoid introducing artificial predictability. Note that "no spurious predictability" is often assessed by a small collection of informal tests whose precise definition may vary across studies.

Aim. Let (r_t) be returns. The reference constructs a synthetic path law by: (i) extracting short representative segments ("motifs") using a time-warping-type similarity, (ii) generating such segments by a conditional diffusion mechanism, and (iii) assembling long paths by a transition rule. The guiding idea of the project is elementary: under a fixed, simple information set, the next return should not exhibit a systematic predictable drift.

Step 1 (Replication): reproduce the paper's construction end-to-end. Reproduce the paper's full three-stage construction as stated: (i) motif extraction (with the same time-warping similarity and clustering choices), (ii) conditional diffusion generation of motif segments, and (iii) assembly into long paths by the paper's transition mechanism. The replication must enforce strict time separation (motif extraction and all fitted objects are trained on training windows only). Reproduce the paper's principal evaluation results on at least one dataset/protocol and implement the baselines used in the paper.

Step 2 (Optional extension): add one fixed no-predictable-drift diagnostic. Keep the generator from Step 1 unchanged. Fix once and for all:

- a rolling time-splitting rule for estimation;
- the feature map, e.g.

$$\phi_t = (1, r_t, r_{t-1}, |r_t|, |r_{t-1}|), \qquad g_b(\mathcal{F}_t) = b^\top \phi_t, \qquad \|b\|_2 \le 1;$$

- the estimator of the drift-violation statistic

$$\Delta(\nu) \;=\; \sup_{\|b\|_2 \le 1} \Big| \mathbb{E}_\nu \big[r_{t+1}\, b^\top \phi_t \big] \Big|,$$

  computed by the fixed rolling scheme.

Compute Δ for (i) real data, (ii) synthetic data from the Step 1 generator, and (iii) a simple "null" constructed from the real data (e.g. block-shuffling returns to break predictability while keeping the marginal distribution approximately intact). Report Δ together with its variability over repeated runs, and report the stylised-fact checks (tails; autocorrelation of r_t and $|r_t|$) alongside it.

Step 3 (Optional headroom): reduce Δ by one penalty and test on a final hold-out. Split data into training / validation / final hold-out (the hold-out is not used for choices). Modify training by adding one scalar penalty weight $\lambda \ge 0$ which discourages predictable drift as measured by the fixed statistic above (same ϕ_t, same rolling scheme). Choose λ on the validation period by a fixed rule stated in advance, then apply the chosen λ unchanged to the final hold-out and (if possible) to a second asset. Report the trade-off between (i) stylised-fact fidelity (tails and volatility clustering) and (ii) the drift statistic Δ. The claim remains empirical: one additional penalty, chosen once, reduces predictable drift under a fixed test without destroying the basic distributional features, and the effect persists out-of-sample.